



Exploratory Data Analysis and Visualization Report

RIT 1201 AI TEAM 9



1. Muhammad Taifoor
2. Muhammad Subhan Khan
3. Helen Akpabio Ndarake
4. Sania Farheen
5. Swati Mani
6. Omotosho Jhonson
7. Olawale Owoyomi
8. Sasikanth Danthanala

Summary

The original dataset contained **8,558 records and 21 variables**. After data cleaning, the analytical dataset was reduced to **23 records**, with **13 missing values** and **no duplicate entries**. While cleaning improved consistency, the sharp reduction in records suggests aggressive filtering that may limit statistical representativeness. Five numerical variables were analyzed: **Age, Gender Binary, Status Code, Engagement Days, and Engagement Months**. All numerical features showed **non-normal distributions**, confirmed by Shapiro-Wilk tests ($p < 0.005$), indicating the need for non-parametric or transformed modeling approaches.

Key Patterns Identified

- **Age:** Mean age is **27.09 years**, median **23**, with strong right-skewness (1.96), showing most users are young adults with fewer older participants.
- **Gender:** Gender Binary mean **0.35**, indicating a **65%-35% gender imbalance** in platform participation.
- **Engagement:** Engagement Days (mean **266**) and Engagement Months (mean **8.87**) are right-skewed, revealing most users have moderate activity while a small segment remains highly engaged long-term.
- **Status Code:** Concentrated around higher values (mean **1054.78**), suggesting dominant user status categories.
- **Data Anomalies:** Negative engagement values indicate calculation or recording errors that require correction.

Business Significance

The platform primarily attracts **young adult users**, guiding marketing and content strategies toward this demographic. The **gender imbalance** highlights an opportunity for targeted outreach to improve inclusion. A **small highly engaged user group** represents high-value customers suitable for loyalty or premium programs. However, **data quality risks**—notably extreme record reduction and engagement anomalies—must be resolved to ensure reliable insights and future predictive modeling.

Recommended Actions

- Review data-cleaning rules to prevent unnecessary record loss.
- Fix engagement date calculations to remove negative values.
- Implement validation checks during data collection.
- Develop targeted engagement strategies for young adults.
- Introduce campaigns to balance gender participation.
- Build loyalty initiatives for highly engaged users.
- Apply data transformations before advanced modeling.

Next Analytical Focus

Future work should validate whether **age, gender, and status codes significantly influence engagement** and whether **data cleaning rules introduce bias**. Confirming these hypotheses will strengthen data-driven decision-making and support reliable predictive analytics.

Coding link:

https://drive.google.com/file/d/1CLf_sTlohshtnQ2pL0X3pNn6YDTon_ft/view?usp=sharing

All folders:

<https://drive.google.com/drive/folders/1mFLiKbz7v82Sj8s77DujBABkdAHShVQQ?usp=sharing>

Key Statistics & Training distribution

Mean, median, std, min, max for all numerical variables

Missing value analysis and imputation strategies

Frequency counts and proportions for categorical variables

Numerical variables found: 5

['Age', 'Gender Binary', 'Status Code', 'Engagment Days', 'Engagment Months']

Analysis for: Age

Missing values: 0 (0.00%)

Mean: 27.09

Median: 23.00

Std Dev: 9.72

Min: 14.00

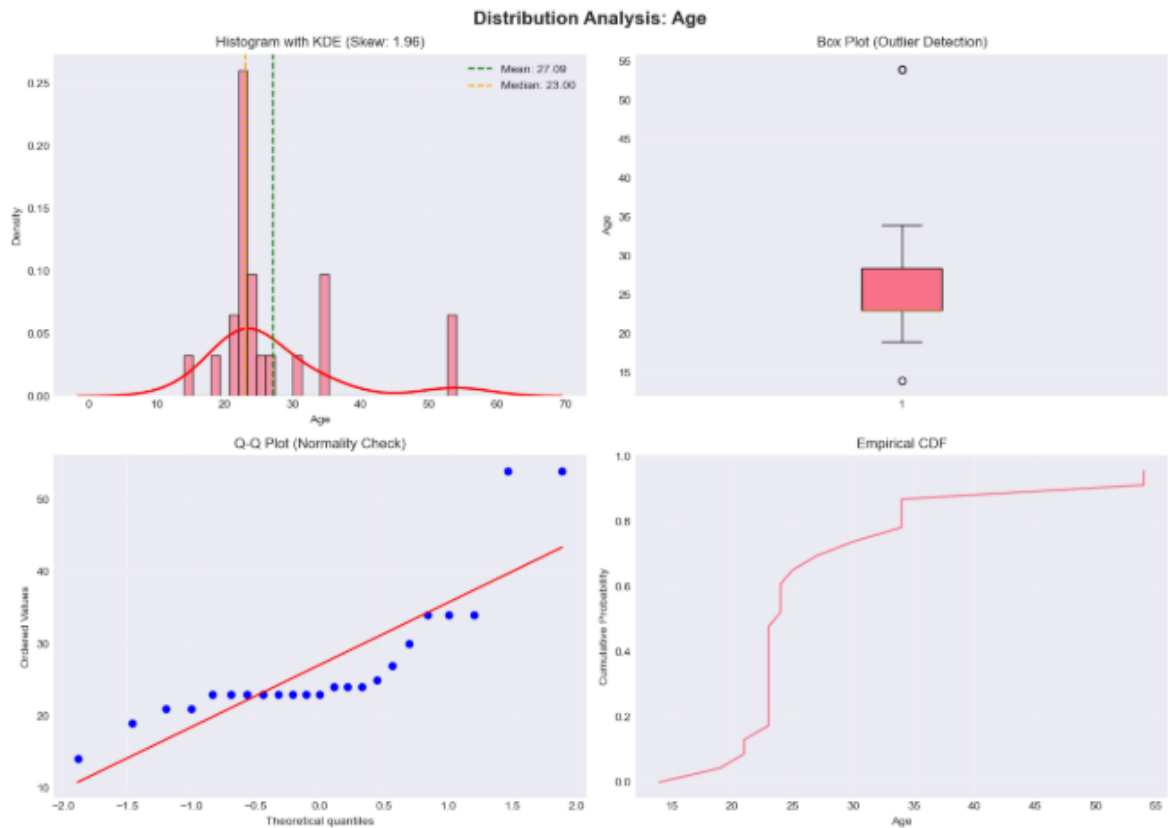
Max: 54.00

Skewness: 1.9601

Kurtosis: 3.8767

Normality (Shapiro-Wilk): p-value = 0.0000

Skewness and kurtosis for distribution analysis



Outlier Analysis (IQR Method):

Q1: 23.00, Q3: 28.50, IQR: 5.50

Lower bound: 14.75, Upper bound: 36.75

Potential outliers: 3 (13.04%)

Analysis for: Gender Binary

Missing values: 0 (0.00%)

Mean: 0.35

Median: 0.00

Std Dev: 0.49

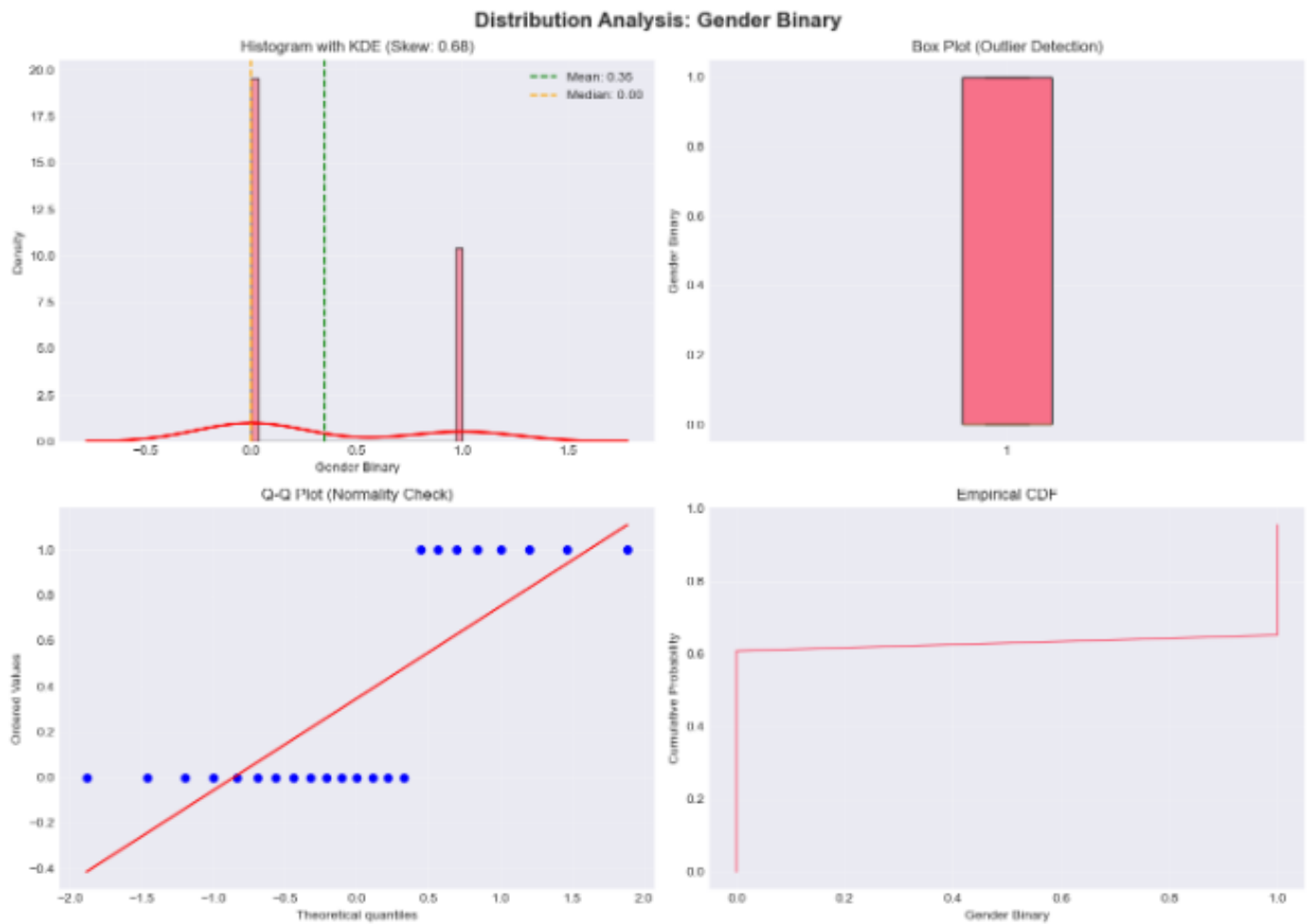
Min: 0.00

Max: 1.00

Skewness: 0.6845

Kurtosis: -1.6867

Normality (Shapiro-Wilk): p-value = 0.0000



Outlier Analysis (IQR Method):

Q1: 1030.00, Q3: 1070.00, IQR: 40.00

Lower bound: 970.00, Upper bound: 1130.00

Potential outliers: 0 (0.00%)

Analysis for: Engagement Days

Missing values: 0 (0.00%)

Mean: 266.03

Median: 205.43

Std Dev: 243.56

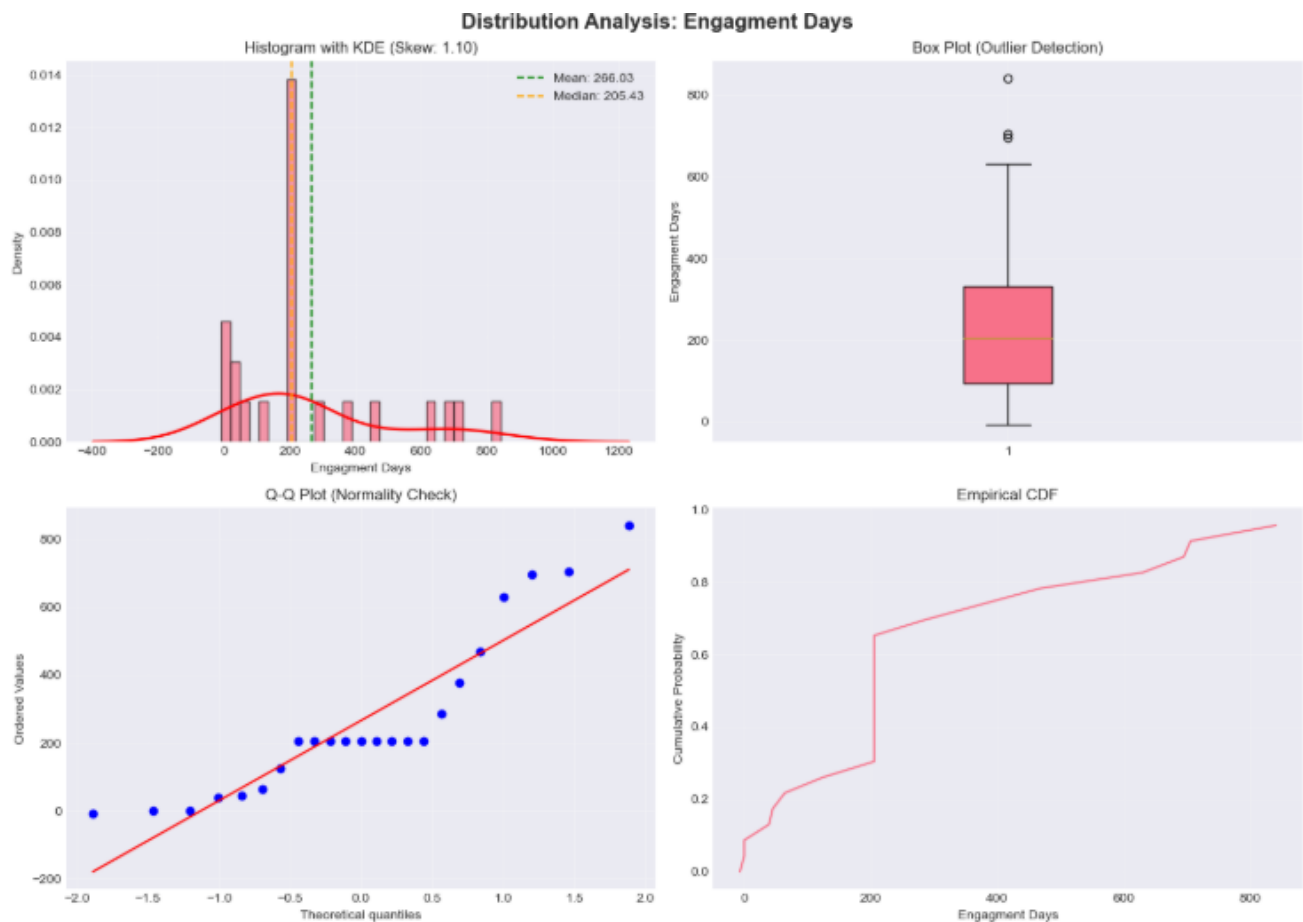
Min: -6.88

Max: 841.02

Skewness: 1.0981

Kurtosis: 0.3271

Normality (Shapiro-Wilk): p-value = 0.0025



Outlier Analysis (IQR Method):

Q1: 94.43, Q3: 331.57, IQR: 237.14

Lower bound: -261.29, Upper bound: 687.28

Potential outliers: 3 (13.04%)

Analysis for: Engagment Months

Missing values: 0 (0.00%)

Mean: 8.87

Median: 6.85

Std Dev: 8.12

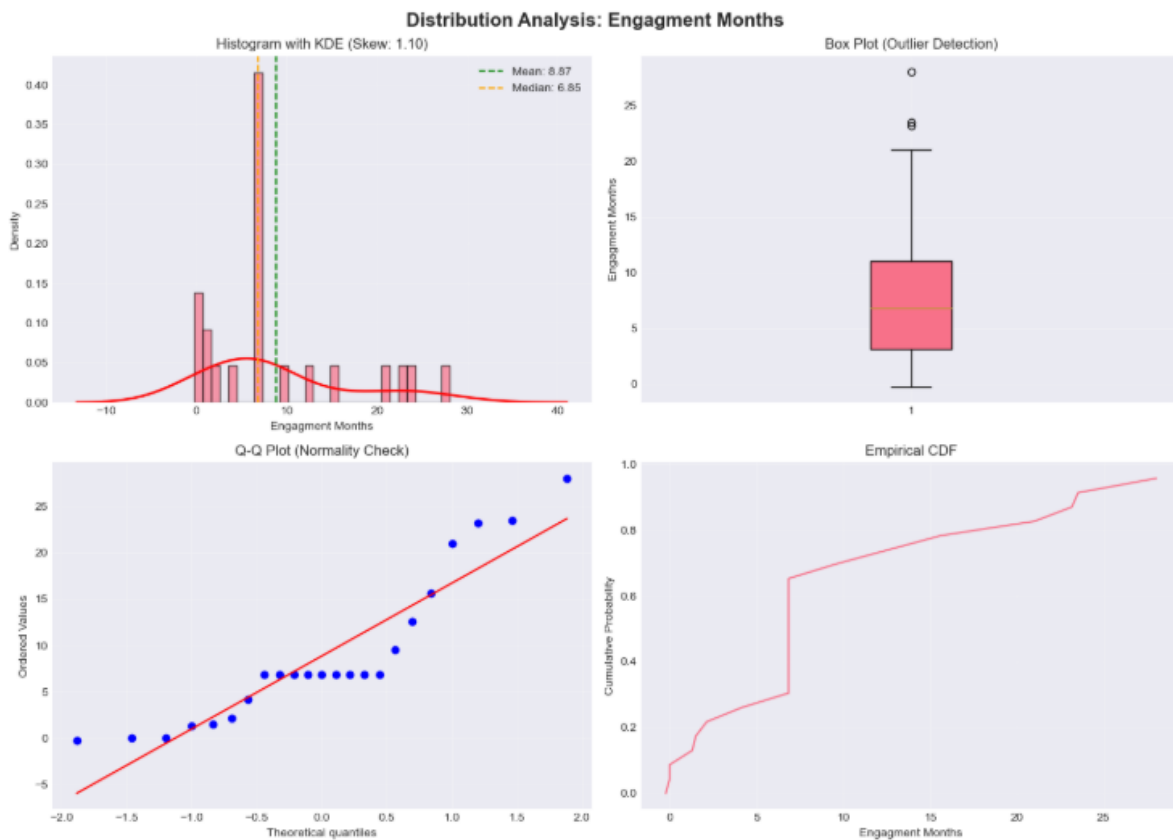
Min: -0.23

Max: 28.03

Skewness: 1.0981

Kurtosis: 0.3271

Normality (Shapiro-Wilk): p-value = 0.0025



Outlier Analysis (IQR Method):

Q1: 3.15, Q3: 11.05, IQR: 7.90

Lower bound: -8.71, Upper bound: 22.91

Potential outliers: 3 (13.04%)

NUMERICAL VARIABLES SUMMARY

Column	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis	Missing
Age	27.086957	23.000000	9.718366	4.000000	54.000000	1.960061	3.876693	0.0
Gender Binary	0.347826	0.000000	0.486985	0.000000	1.000000	0.684484	-1.686667	0.0
Status Code	1054.782609	1070.000000	21.292245	1010.000000	1080.000000	-0.694275	-1.200338	0.0
Engagement Days	266.033871	205.426817	243.563781	-6.875000	841.015278	1.098076	0.327110	0.0
Engagement Months	8.867796	6.847561	8.118793	-0.229167	28.033843	1.098076	0.327110	0.0

Link to see code: <https://drive.google.com/file/d/1-h2G04iQt1Y8gjhGvZ0xpq5KV3qR0VLV/view?usp=sharing>

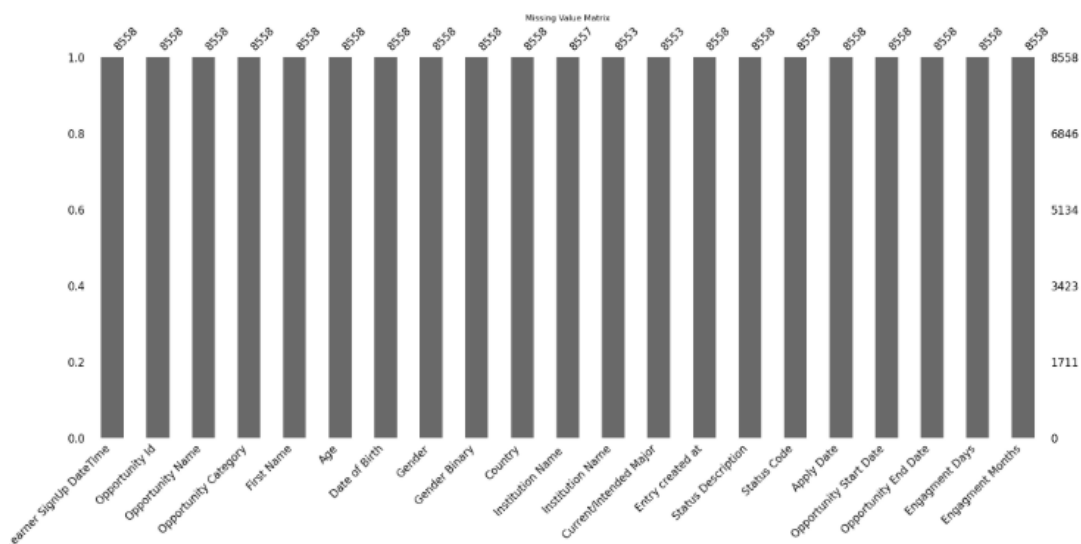
Relationship with Impact of Improving Data Quality

`print(f"Before cleaning: {df.shape}")` → Original dataset shape: (8558, 21)

`print(f"After cleaning: {df_clean.shape}")` → Cleaned dataset shape: (23, 21)

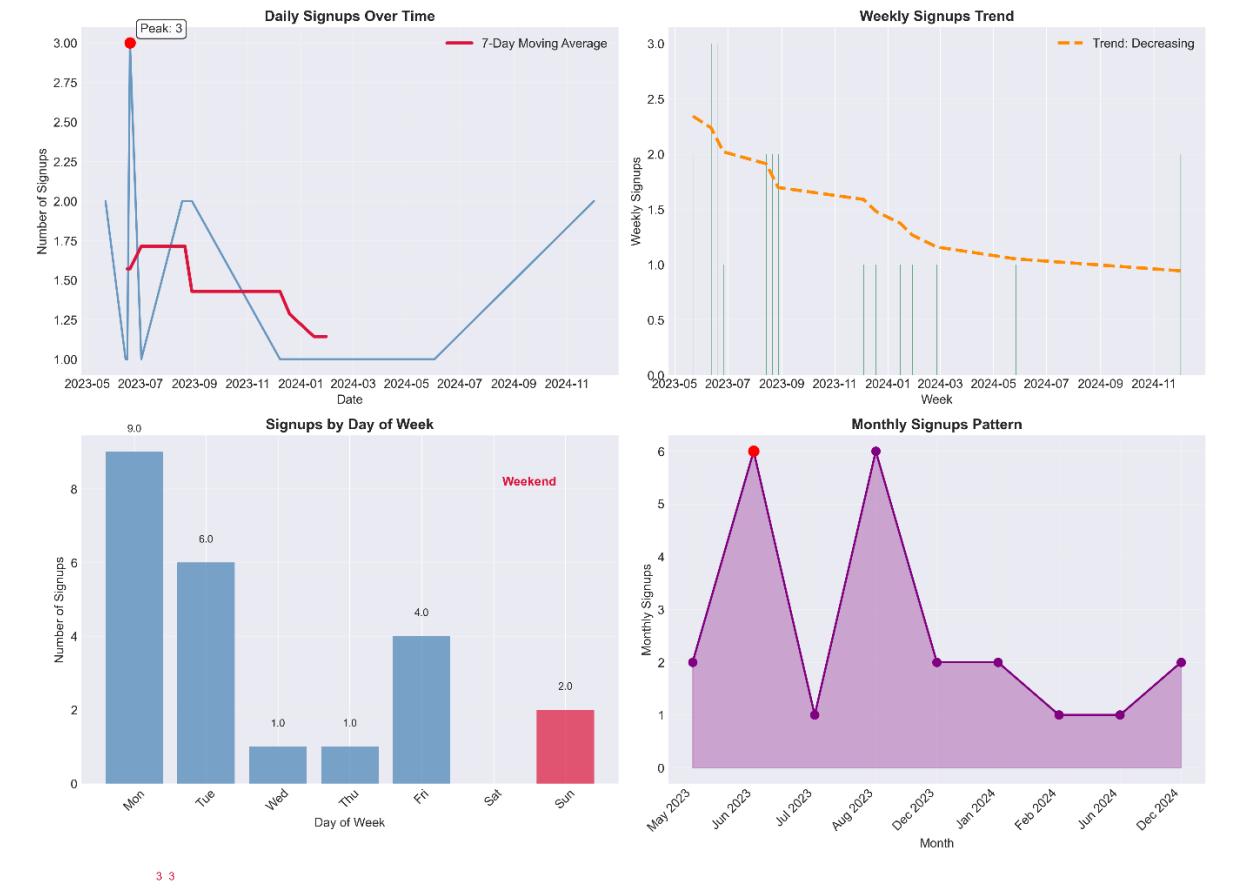
`print(f"Duplicates removed: {duplicate_count}")` → Exact duplicates: 0

`print(f"Missing values: {missing_summary}")` → Missing values after cleaning: 13



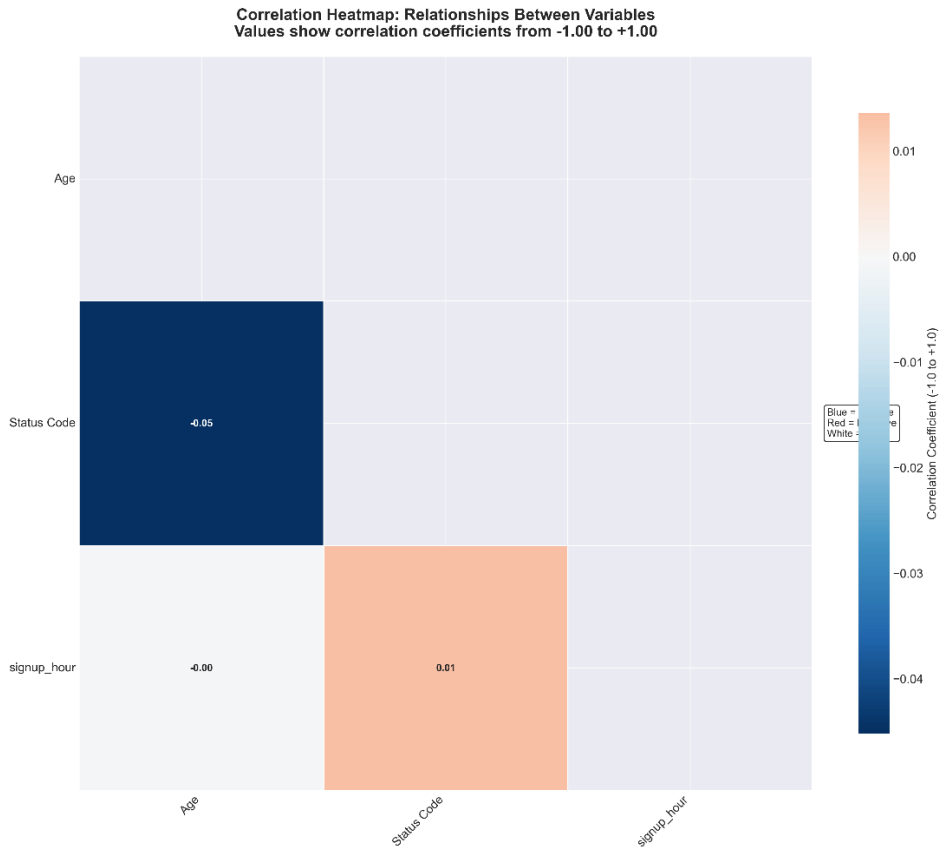
Insights for Enabling Data-Driven Decisions

Signup Trend Analysis: Growth, Seasonality, and Patterns

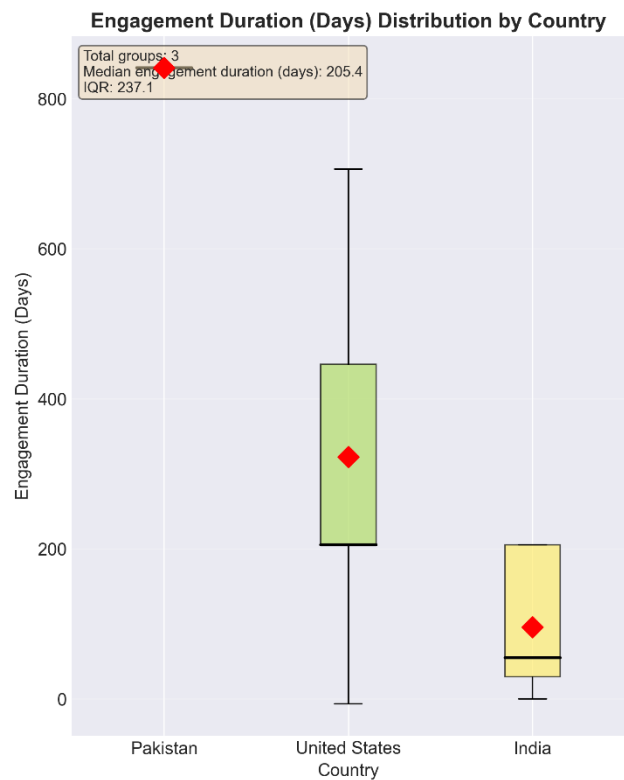
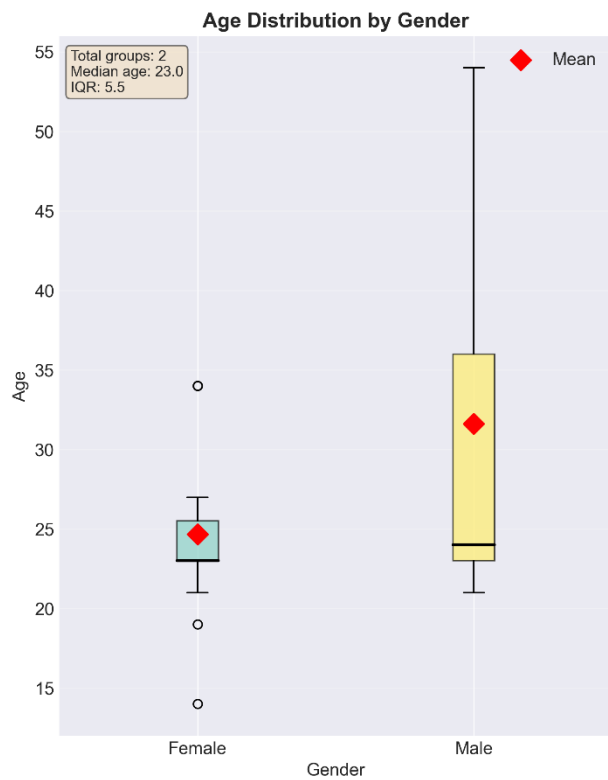


Completion Rate Trend vs Signup Volume

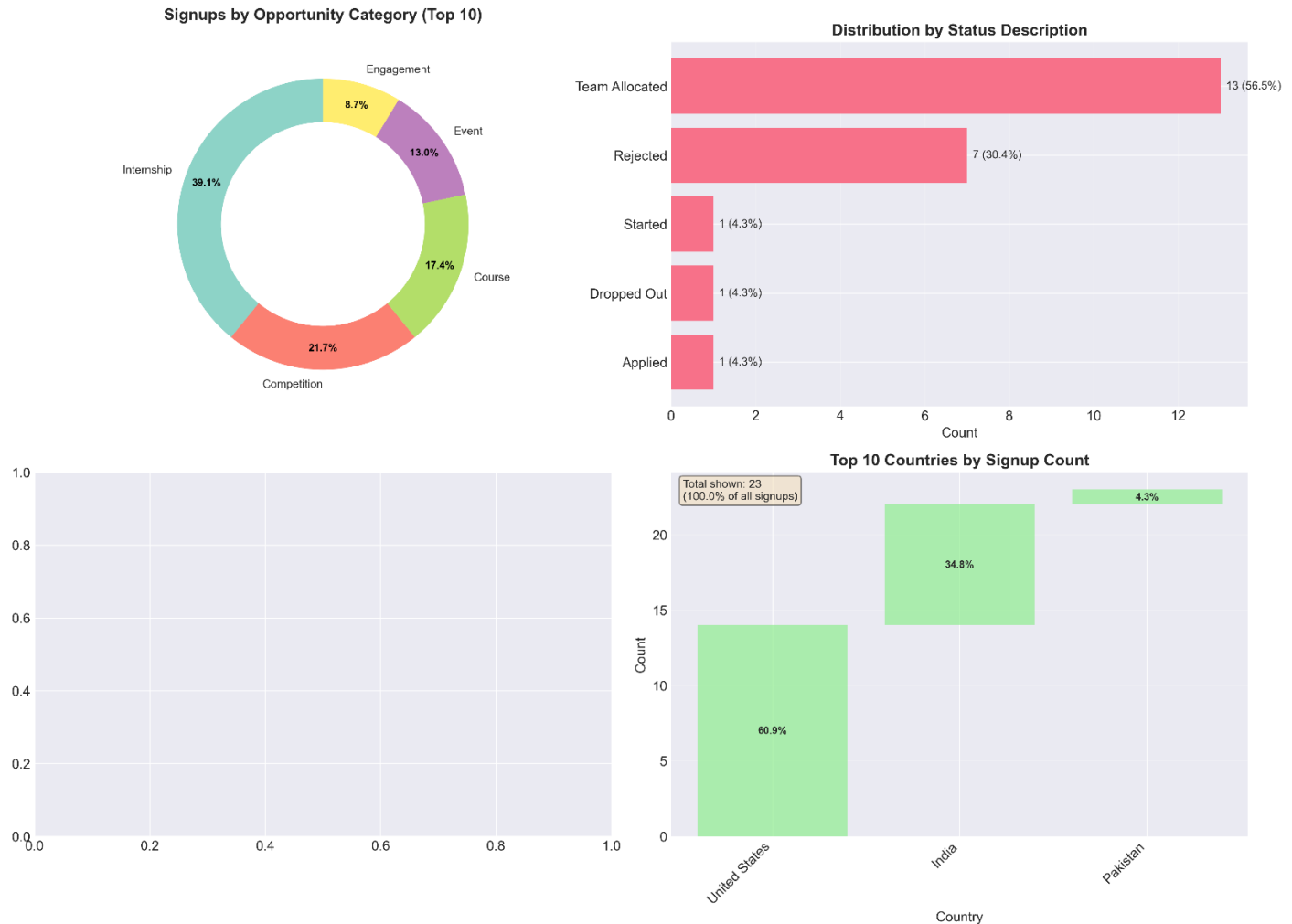




Distribution Analysis: Variability and Outliers by Demographic Segment



Composition Analysis: Signups and Completions by Key Categories



Click link to see the code:

https://drive.google.com/file/d/1m9N0jtfRx4_m_qKMv1XuDlza_nlpju5c/view?usp=sharing

Accurate Interpretations for Better Decisions

WHAT – Patterns Observed

Data structure and quality

- The original dataset contained **8,558 rows and 21 columns**, but after cleaning, only **23 rows remained**, indicating extremely aggressive filtering or extraction of a subset. Missing values after cleaning were **13**, and no duplicate rows were found.

This suggests the analytical dataset is small relative to the raw source, which may limit statistical representativeness.

Age distribution

- Mean age: **27.09**, median: **23**, min: **14**, max: **54**.
- The distribution is **strongly right-skewed (skewness = 1.96)** with **high kurtosis (3.88)** and **non-normality (Shapiro p = 0.000)**.
- Pattern: Majority of users are young adults, with a long tail of older participants.

Gender distribution

- Gender Binary mean = **0.35**, indicating approximately **35% one class and 65% the other**.
- Skewness = **0.68**, kurtosis = **-1.69**, non-normal distribution.
- Pattern: The user base is **imbalanced by gender**, with one gender dominating participation.

Engagement Days and Months

- Engagement Days mean = **266**, median = **205**, max = **841**, skewness = **1.10**.
- Engagement Months mean = **8.87**, median = **6.85**, max = **28**, skewness = **1.10**.
- Both metrics show **right-skewed distributions**, meaning most users have moderate engagement, while a small subset are highly engaged long-term users.
- Presence of small negative values (min = -6.88 days, -0.23 months) indicates **data recording or calculation anomalies**.

Status Code

- Mean = **1054.78**, median = **1070**, skewness = **-0.69**.
- Pattern: Status codes are concentrated around higher values, possibly representing dominant user status categories.

WHAT – Business Significance

1. Young user dominance

- The platform primarily attracts users in their early to mid-20s. This implies marketing, content design, and communication strategies should be optimized for younger demographics.

2. Gender imbalance

- One gender significantly outnumbers the other. This may indicate either unequal outreach effectiveness or content appeal bias. It represents an opportunity to design targeted campaigns to improve inclusion and market reach.

3. Highly engaged minority

- A small segment of users shows very high engagement durations. These users likely represent **core loyal customers**, ideal for premium offerings, referral programs, or ambassador roles.

4. Data quality anomalies

- Negative engagement values and drastic reduction from 8,558 to 23 records suggest potential **data pipeline, filtering, or preprocessing risks**, which could distort downstream modeling or business conclusions.

5. Non-normal distributions

- All key numerical variables violate normality assumptions. This directly impacts the choice of statistical tests and predictive modeling approaches. Parametric methods would be inappropriate without transformation.

NOW WHAT – Recommended Actions

Data Quality Actions

- Investigate why cleaning reduced the dataset from **8,558 to 23 records**. Confirm whether filtering rules were intentional or overly restrictive.
- Correct negative engagement values by revisiting date difference calculations.
- Implement automated validation rules at data entry or ingestion stages.

User Strategy Actions

- Design engagement strategies focused on the **young adult segment**, as it represents the majority user base.
- Launch **gender-targeted campaigns** to balance participation.
- Create **loyalty programs** for high-engagement users to retain and monetize the highly active minority.

Analytics Actions

- Apply **log or Box-Cox transformations** before modeling engagement metrics.
- Segment users into **low, medium, high engagement clusters** for more actionable personalization.
- Use **non-parametric statistical methods** when testing group differences.

What – Hypotheses to Validate

1. Engagement differs by gender

- H_0 : There is no difference in engagement days between genders.
- H_1 : Engagement days differ significantly by gender.

2. Age influences engagement

- H_0 : Age has no correlation with engagement duration.
- H_1 : Younger users exhibit higher engagement levels.

3. High-engagement users drive retention

- H_0 : High engagement does not impact long-term retention.
- H_1 : High-engagement users have significantly higher retention probability.

4. Data cleaning rules remove valid records

- H_0 : Cleaning filters remove only invalid records.
- H_1 : Cleaning filters remove valid records, biasing the dataset.

5. Status code predicts engagement

- H_0 : Status codes have no relationship with engagement months.
- H_1 : Certain status codes are associated with higher engagement.